Polar and hexagonal maze profiles

Created-at: 2026-10-06 14:12 CEST. ETA: 50–70 minutes: profile geometry and canonical validation, owner tests, Amaterasu generation and physical handoff.

Owner Formats composes maze descriptions; Core owns IR and semantic identity; Query reads authored cells/walls; Graph owns weighted spanning-tree selection and shortest paths; Geometry owns capsule/box collision checks. No new parser, graph engine, geometry engine or renderer. A separate collaborator owns physical model composition and the existing MuJoCo/Kawamirai execution.

Profiles First delivery is a nineteen-cell axial hex maze (radius2; radius1 remains a small topology test) and a two-ring, six-sector polar maze. Seeded Graph LCG weights and rank-2 matroid basis produce connected acyclic passage systems. Hex has six-neighbor topology; polar has angular and radial neighbors. Explicit boundary segments generate oriented boxes; circular walls use three facets per sector. The common export DTO contains cell positions, passages, yaw boxes, endpoints and route points.

Correctness Canonical source roundtrip and semantic digest; Query counts and finite field validation; exact geometry/topology consistency; Graph BFS/Dijkstra distance agreement; seed replay and variation; no passage through a closed wall. Polar angular moves use intermediate arc points. Geometry's capsule/box query checks the full route against a 0.85m clearance radius. This is a geometric corridor witness, not controller success.

Files Existing example main and visual module; new `hymeko_formats/examples/adversarial_grid/maze.rs`; focused tests in that module; README; this plan and local report/evidence. The collaborator owns `adversarial_grid/maze_physics.rs`; no overlapping edits there or in physical.rs. CORE.YAML touched: []. No dependency, public API, schema of old grid or old physical parameters changed. New profile schema is explicit and versioned.

Bounds At most 19 profile cells, 72 uncut wall segments, bounded Graph work. No large campaign or benchmark. Two generated examples, seeds specified, run on Amaterasu only with fresh output and timeout 180s. Unit tests are pure and may run locally. Two build jobs; no install, commit/push/merge, new worktree, Mission Control changes, or implicit simulation.

Risks Polar chords can cut corners; test full route capsule against actual faceted walls. Arc joins need overlap. RNG yields a randomized spanning tree, not a claim of uniformly sampled trees. A provenance mismatch or clearance failure refuses export. Preserve existing dirty owner work; rollback removes only this reviewed continuation's delta and leaves prior artifacts unchanged.

